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Title : JMeter Performance Testing & CI/CD Integration

1) Create the JMeter Test Plan : Create a JMeter test plan containing five Thread Groups.

2) Configure the Thread Groups

Configure each Thread Group with:

- Number of Users: **10**
- Ramp-Up Period: **5 seconds**
- Loop Count: **3**

3) Configure Test Scenarios: Add the required HTTP Request samplers, test data, parameters, listeners and assertions for each application also

4) Execute the Test in Non-GUI Mode: Execute the JMeter test using the command line to perform the load test efficiently.

5) Analyze Results Using Aggregate Graph/Report

Open the generated JTL file in JMeter and analyze the performance results using the Aggregate Graph/Report.

6) Validate Performance Metrics

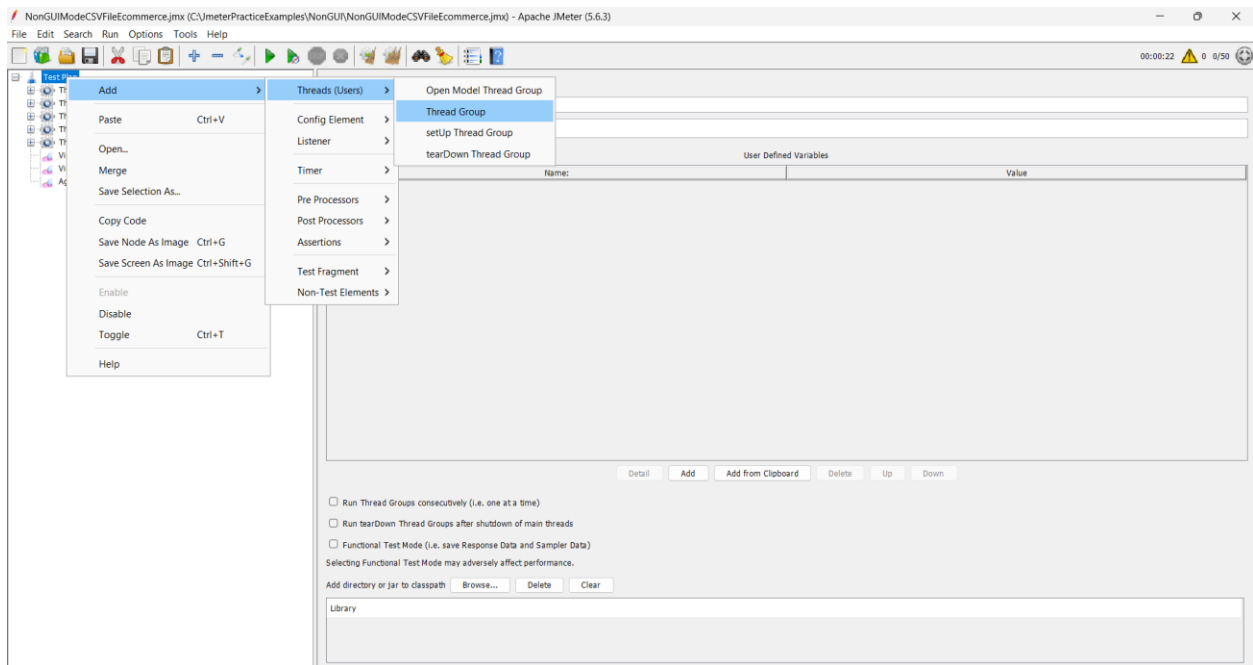
- Total Samples
- Throughput
- Latency
- Median
- Minimum Response Time
- Maximum Response Time
- 90th Percentile
- 95th Percentile
- Error Percentage

7) Generate the HTML Dashboard Report and Integrate with jenkins

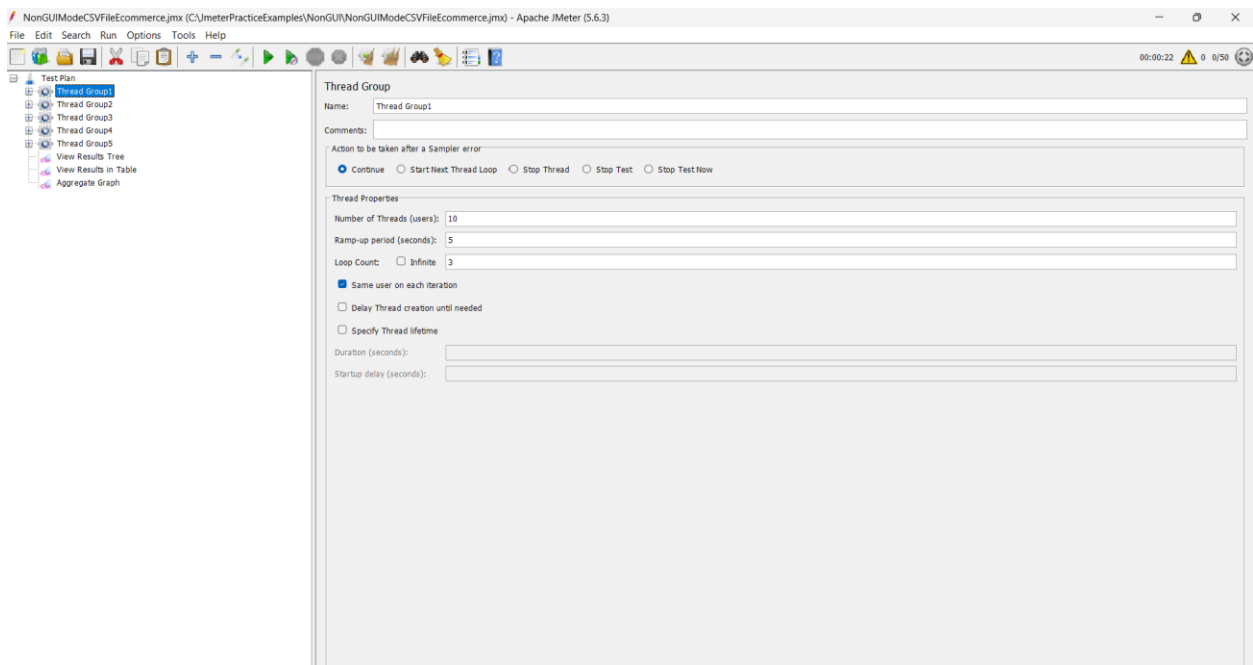
Use the JTL file to generate the JMeter **HTML Dashboard Report** containing graphical performance statistics.

8) Validate Jenkins Execution

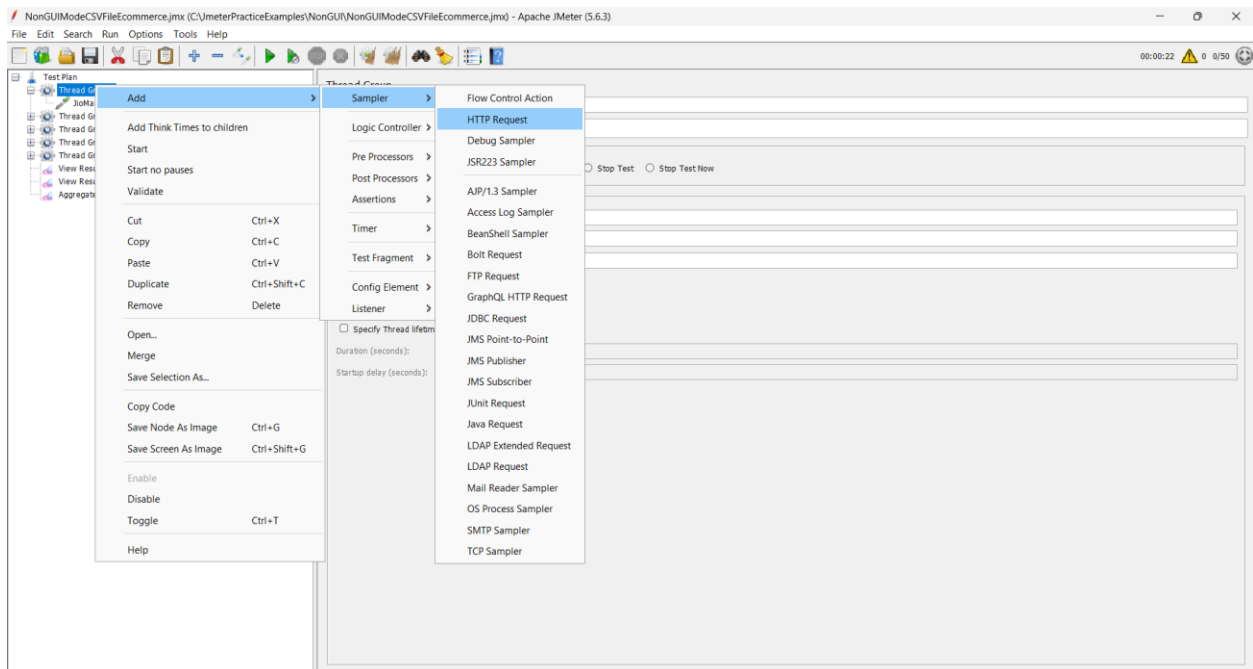
Step 1: Create a JMeter test plan containing five Thread Groups



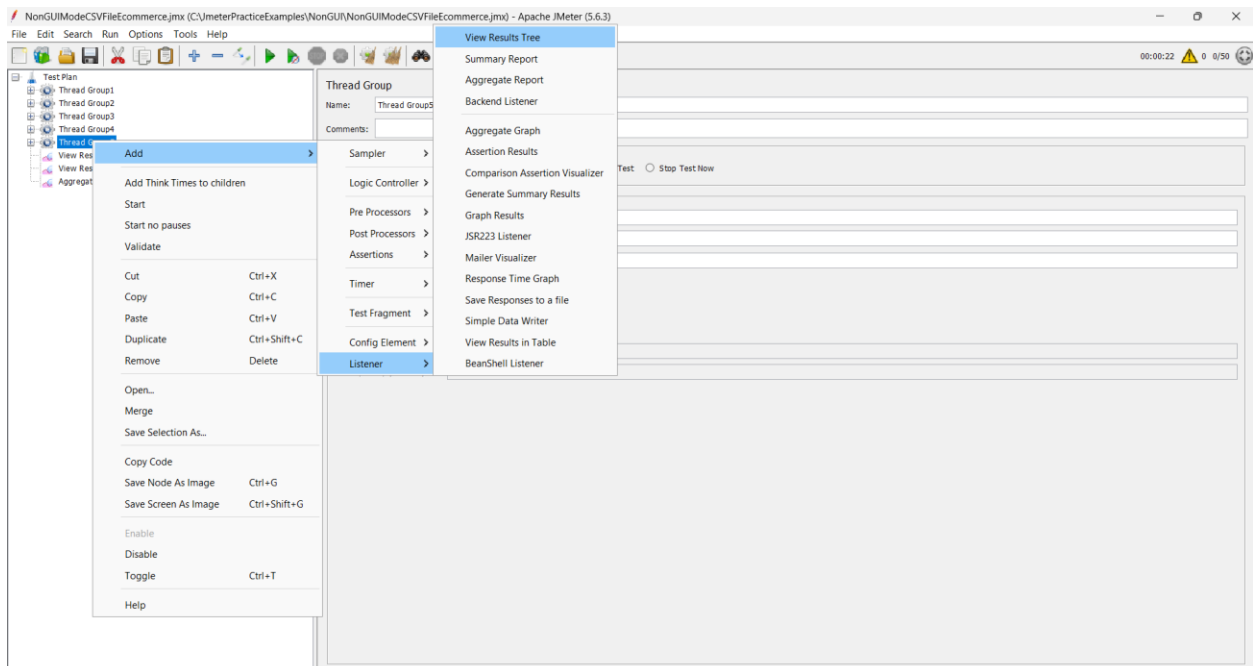
Step 2: Configure each Thread Group according to the requirements mentioned

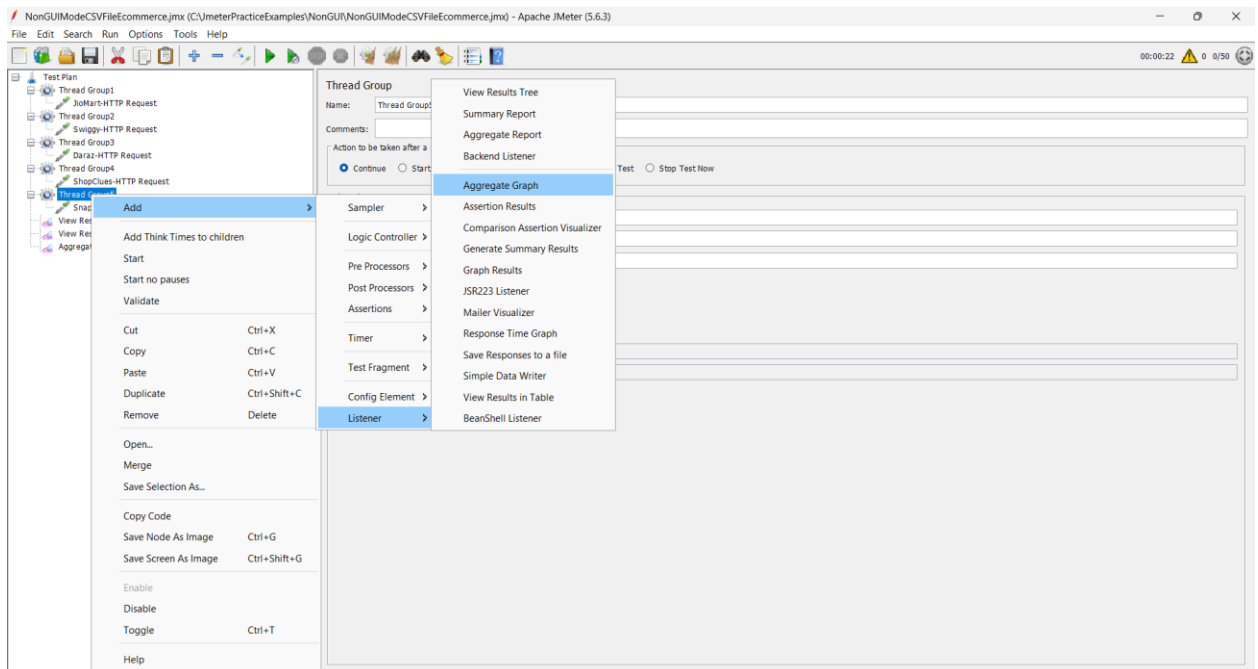
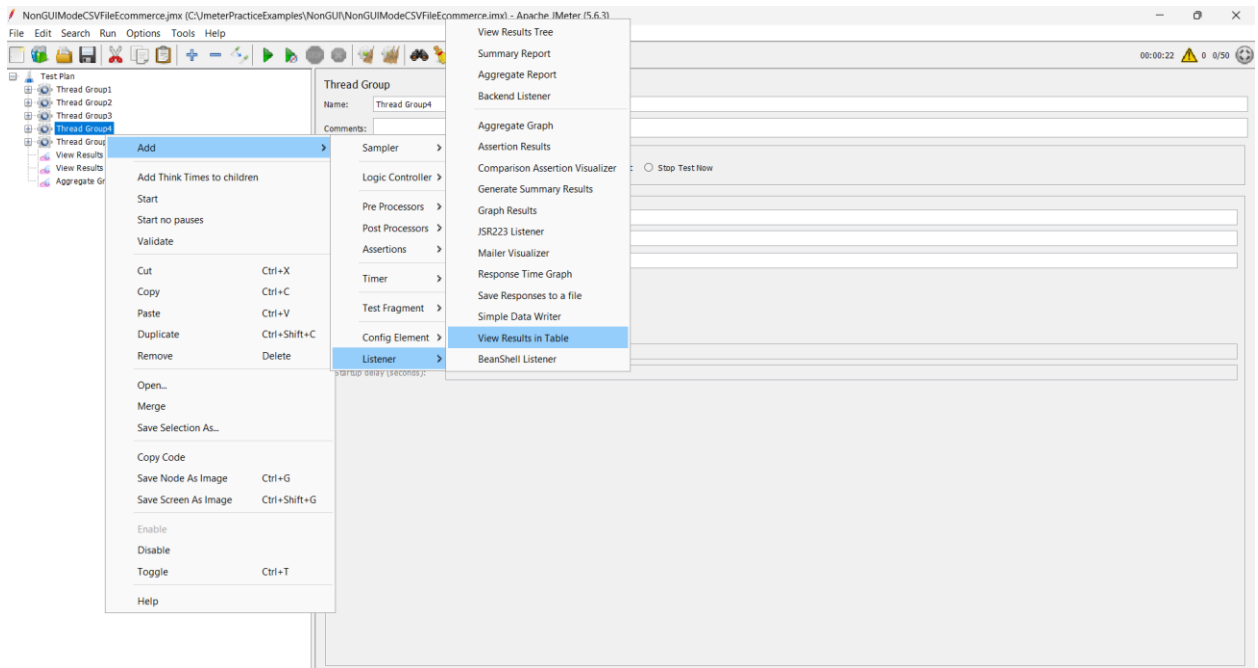


Step 3: Add samplers to each thread group using different E-commerce websites



Step 4: Add listeners





Step 5 : Run and validate

NonGuiModeCSVFileCommerceExamples\CGMeterPracticeExamples\NGUI\NonGuiModeCSVFileCommerceExamples - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:22 ⚠️ 0/0/50

Test Plan

- Thread Group
 - 3rdParty HTTP Request
- Thread Group2
 - Swingy HTTP Request
- Thread Group3
 - Daraz HTTP Request
- Thread Group4
 - ShopClues HTTP Request
- Thread Group5
 - SnapDeal HTTP Request

[View Results Tree](#)
[View Results in Table](#)
[Aggregate Graph](#)

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file
Filename: Browse... LogDisplay Only: ☐ Errors ☐ Successes ☒ Configure

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	18:19:03.410	Thread Group4 4-1	ShopClues-HTTP Req...	789	✔️	316269	143	712	669
2	18:19:03.746	Thread Group4 4-2	ShopClues-HTTP Req...	436	✔️	316270	143	354	312
3	18:19:04.281	Thread Group4 4-1	ShopClues-HTTP Req...	48	✔️	316268	143	0	143
4	18:19:04.256	Thread Group4 4-1	ShopClues-HTTP Req...	436	✔️	316268	143	0	0
5	18:19:04.205	Thread Group4 4-2	ShopClues-HTTP Req...	110	✔️	316267	143	34	0
6	18:19:04.316	Thread Group4 4-2	ShopClues-HTTP Req...	78	✔️	316268	143	25	0
7	18:19:03.410	Thread Group5 5-1	SnapDeal-HTTP Req...	1063	✔️	386397	154	1016	647
8	18:19:04.268	Thread Group5 4-3	ShopClues-HTTP Req...	211	✔️	316267	143	136	94
9	18:19:04.474	Thread Group5 5-1	SnapDeal-HTTP Req...	106	✔️	386291	154	124	0
10	18:19:04.480	Thread Group4 4-3	ShopClues-HTTP Req...	124	✔️	316267	143	54	0
11	18:19:04.580	Thread Group5 5-1	SnapDeal-HTTP Req...	111	✔️	385921	154	41	0
12	18:19:04.605	Thread Group4 4-3	ShopClues-HTTP Req...	119	✔️	316267	143	47	0
13	18:19:03.767	Thread Group5 5-2	SnapDeal-HTTP Req...	1218	✔️	386078	154	623	290
14	18:19:04.269	Thread Group5 5-3	SnapDeal-HTTP Req...	734	✔️	386115	154	515	55
15	18:19:03.764	Thread Group3 3-3	Daraz HTTP Request	1335	✔️	228574	311	443	346
16	18:19:04.766	Thread Group5 5-4	SnapDeal-HTTP Req...	390	✔️	385859	154	162	111
17	18:19:03.410	Thread Group3 3-1	Daraz HTTP Request	1772	✔️	228605	311	798	700
18	18:19:05.003	Thread Group5 5-3	SnapDeal-HTTP Req...	211	✔️	385859	154	69	0
19	18:19:05.156	Thread Group5 5-4	SnapDeal-HTTP Req...	216	✔️	385967	154	62	0
20	18:19:04.765	Thread Group4 4-4	ShopClues-HTTP Req...	635	✔️	316267	143	329	196
21	18:19:05.214	Thread Group5 5-3	SnapDeal-HTTP Req...	234	✔️	385967	154	97	0
22	18:19:05.372	Thread Group5 5-4	SnapDeal-HTTP Req...	244	✔️	385967	154	101	0
23	18:19:04.985	Thread Group5 5-2	SnapDeal-HTTP Req...	713	✔️	385959	154	75	0
24	18:19:05.406	Thread Group4 4-4	ShopClues-HTTP Req...	351	✔️	316267	143	134	0
25	18:19:04.266	Thread Group3 3-3	Daraz HTTP Request	1649	✔️	228610	311	362	171
26	18:19:05.793	Thread Group4 4-4	ShopClues-HTTP Req...	271	✔️	316268	143	113	0
27	18:19:05.268	Thread Group5 5-5	SnapDeal-HTTP Req...	780	✔️	385967	154	348	213
28	18:19:03.761	Thread Group2 2-2	Swingy HTTP Request	2395	✔️	355213	126	1921	296
29	18:19:06.048	Thread Group5 5-5	SnapDeal-HTTP Req...	329	✔️	385967	154	145	0
30	18:19:05.267	Thread Group4 4-5	ShopClues-HTTP Req...	1234	✔️	316268	143	581	365
31	18:19:05.099	Thread Group3 3-2	Daraz HTTP Request	1411	✔️	228599	311	169	0
32	18:19:05.182	Thread Group3 3-1	Daraz HTTP Request	1470	✔️	228603	311	183	0
33	18:19:06.378	Thread Group5 5-5	SnapDeal-HTTP Req...	286	✔️	385967	154	185	0

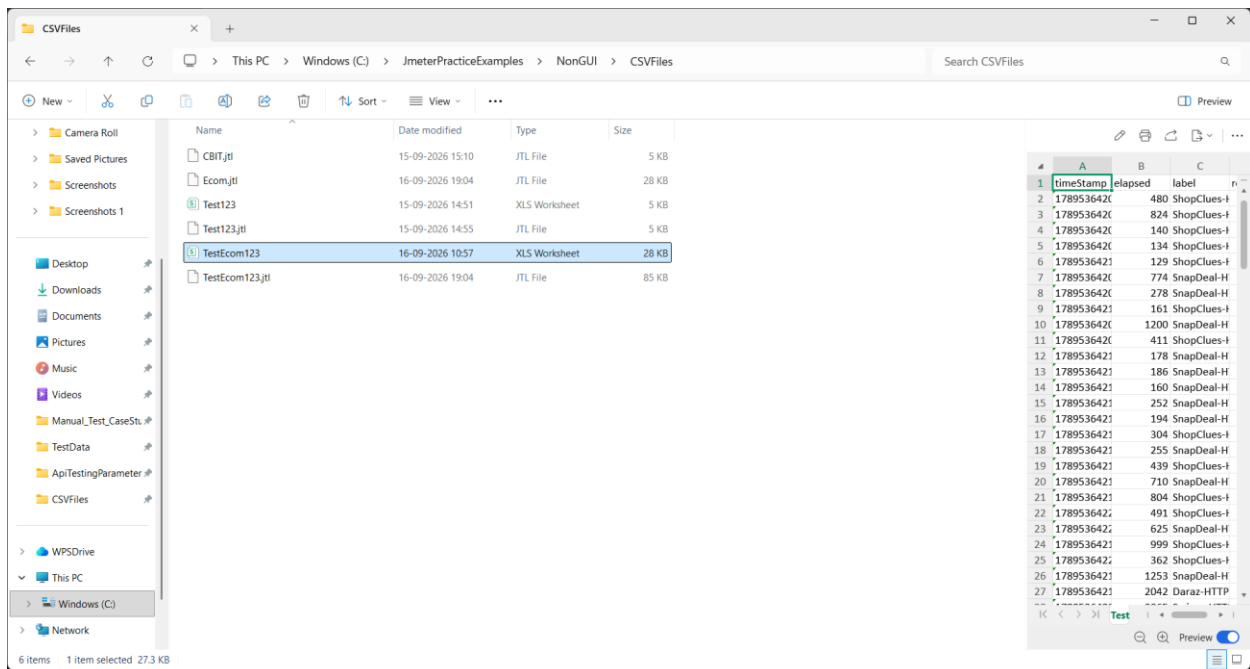
☐ Scroll automatically? ☐ Child samples? It's of Samples 150 Latest Sample 1083 Average 2924 Deviation 3018

The screenshot displays the Apache JMeter GUI. The top menu bar includes File, Edit, Search, Run, Options, Tools, and Help. The top status bar shows the time as 00:00:22 and the progress as 0/50. The left sidebar shows the Test Plan tree, which includes Thread Group1, Thread Group2, Thread Group3, Thread Group4, Thread Group5, and View Results Tree. The View Results Tree is currently selected. The main window shows the View Results Tree panel, which displays a list of test results for various HTTP requests. The results are displayed in a table with columns for Test and Sampler result. The table lists 25 test results, all of which are successful (indicated by green checkmarks). The requests include shopClues-HTTP Request, SnapDeal-HTTP Request, and Daraz-HTTP Request. The bottom status bar shows the time as 00:00:22 and the progress as 0/50.

Step 6:

1. Open Command Prompt.
2. Navigate to the JMeter bin folder using the following command:
3. `cd /d "C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin"`
4. Generate the CSV result file by executing the JMeter test plan in non-GUI mode:
`jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.csv"`

The test execution results are saved in the specified location as a CSV file.

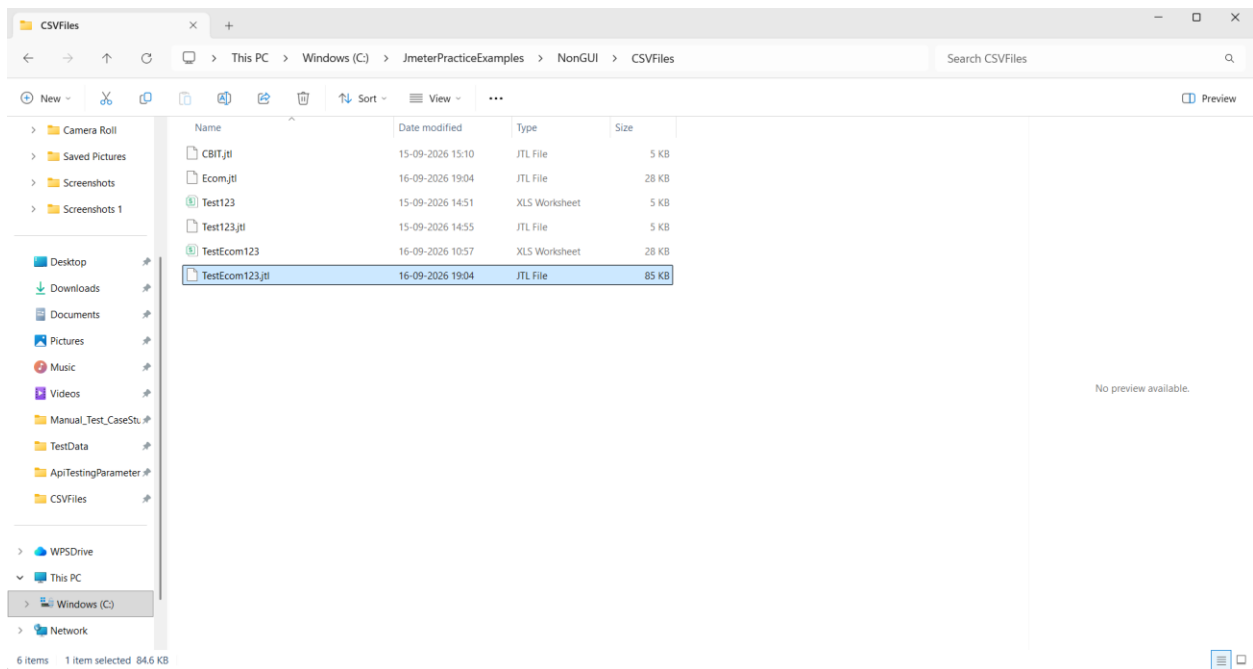


The screenshot shows a Windows File Explorer window titled 'CSVFiles'. The address bar indicates the path: 'This PC > Windows (C:) > JmeterPracticeExamples > NonGUI > CSVFiles'. The file list shows several files, with 'TestEcom123.csv' selected. The preview pane on the right displays the CSV data with the following columns: timestamp, elapsed, label, and n.

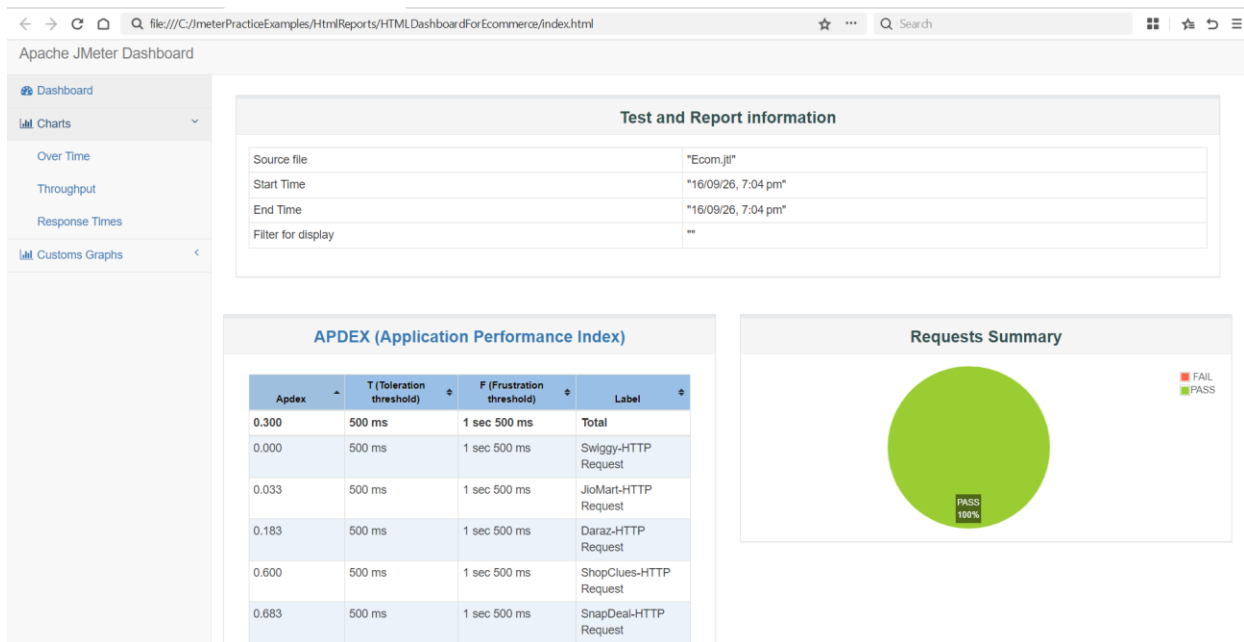
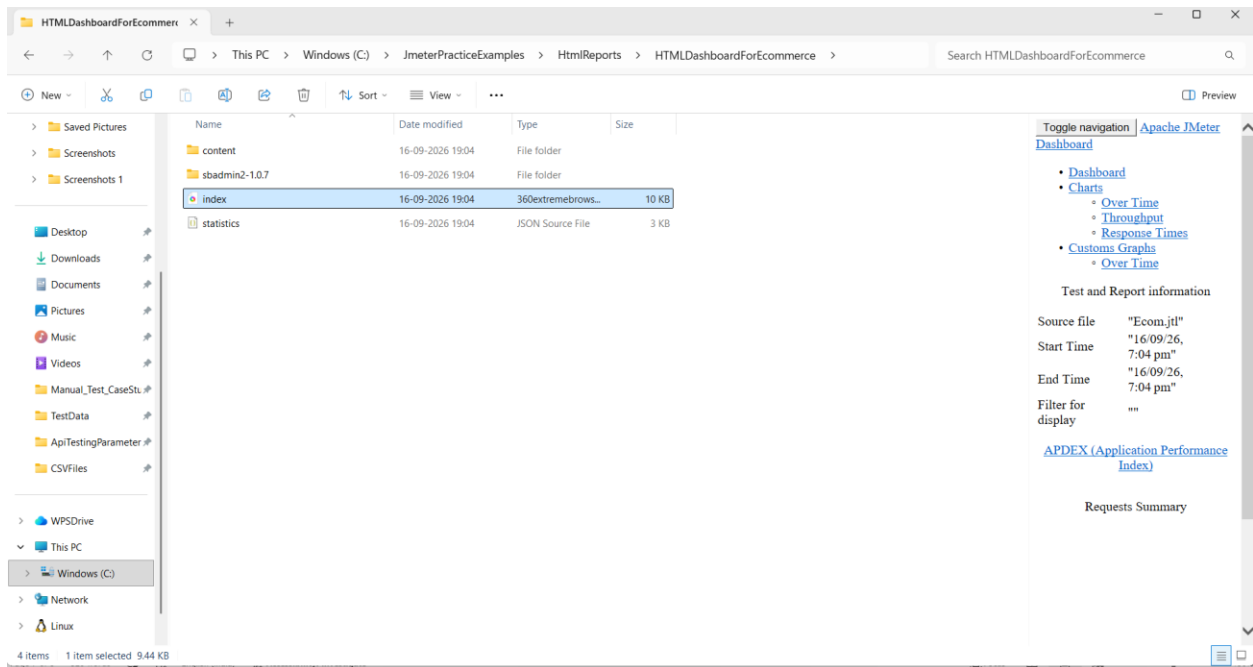
#	A	B	C	n
1	timestamp	elapsed	label	n
2	1789536420	480	ShopClues-H	
3	1789536420	824	ShopClues-H	
4	1789536420	140	ShopClues-H	
5	1789536420	134	ShopClues-H	
6	1789536420	129	ShopClues-H	
7	1789536420	774	SnapDeal-H	
8	1789536420	278	SnapDeal-H	
9	1789536420	161	ShopClues-H	
10	1789536420	1200	SnapDeal-H	
11	1789536420	411	ShopClues-H	
12	1789536420	178	SnapDeal-H	
13	1789536420	186	SnapDeal-H	
14	1789536420	160	SnapDeal-H	
15	1789536420	252	SnapDeal-H	
16	1789536420	194	SnapDeal-H	
17	1789536420	304	ShopClues-H	
18	1789536420	255	SnapDeal-H	
19	1789536420	439	ShopClues-H	
20	1789536420	710	SnapDeal-H	
21	1789536420	804	ShopClues-H	
22	1789536420	491	ShopClues-H	
23	1789536420	625	SnapDeal-H	
24	1789536420	999	ShopClues-H	
25	1789536420	362	ShopClues-H	
26	1789536420	1253	SnapDeal-H	
27	1789536420	2042	Daraz-HTTP	

Step 7:

1. Generate the JTL file to use as the input for the Aggregate Graph: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.jtl"`
2. Open the generated JTL file in JMeter and use the Aggregate Graph/Report to analyze the performance results, including Throughput, Latency, Median, Minimum, Maximum, Percentiles, Total Samples, and Error Percentage.



Step 7: Html reports



Step 9 : Configure Jenkins

Jenkins

JenkinsCICDUsingJmeter

Configure

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Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

cd C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin
jmeter -njmeter.save.saveservice.output_format=xml -n -t
"C:\JmeterPracticeExamples\CICDPracticeTests\NonGUIModeCSVFileEcommerce.jmx"
-l "C:\JmeterPracticeExamples\CICDPracticeTests\JenkinsCICD.jtl"

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Publish Performance test result report

Save

Apply

Step 10: Analyze the performance Trend

Jenkins

JenkinsCICDUsingJmeter

Performance Trend

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Status

Changes

Workspace

Build Now

Delete Project

Performance Trend

Configure

Rename

Test Results Analyzer

Builds >

Filter

Today

#3 1:54 pm

#2 11:47 am

#1 11:44 am

Performance Trend

Last Report

Filter trend data

Test file: JenkinsCICD.jtl

Response time

Percentage of errors

Trend report

Jenkins 2.568.2



Jenkins

JenkinsCICDUsingJmeter

Performance Trend

Trend report



Trend report: JenkinsCICD.jtl

Build	Line 90.0(ms)	Line 95.0(ms)	Max(ms)	Median(ms)	Min(ms)	Percentage of errors	average	average kb	errors	total kb
#2	26267	65067	179643	4495	357	1	11833	1606.88	1	241031.39
#3	19997	42468	179643	4923	228	0	10139	1608.68	1	482603.23

Jenkins 2.568.2

Step 11: JMeter Aggregate Graph – Performance Test Results

